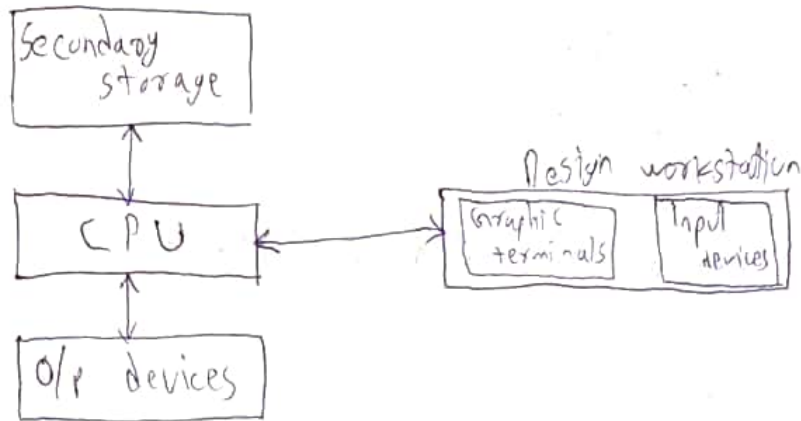


commands to the computer through several i/p devices called designers.

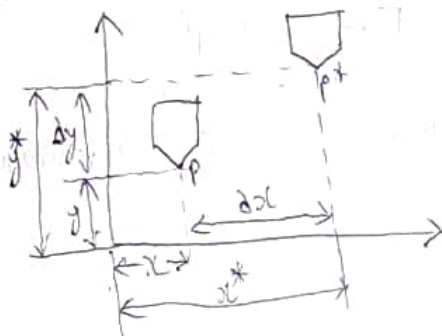
A typical ICG system is a combination of hardware and software.



Hardware components of ICG

Geometric Transformation

1. Translation



Initial position $P = [x, y]$

New position $P^* = [x^*, y^*]$

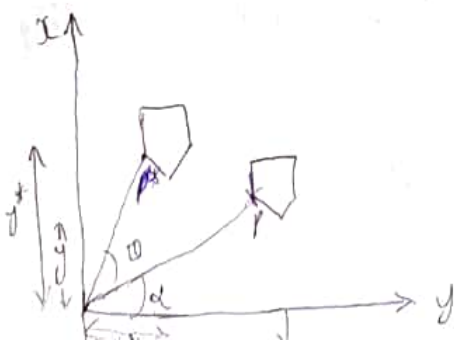
$$[P^*] = \begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} x + dx \\ y + dy \end{bmatrix}$$

$$= \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} dx \\ dy \end{bmatrix}$$

$$[P^*] = [P] + [T_t]$$

$[T_t] \Rightarrow$ Transformation matrix for translation.

2. Rotation



$$x = r \cos \alpha$$

$$y = r \sin \alpha$$

$$x^* = r \cos(\alpha + \theta)$$

$$y^* = r \sin(\alpha + \theta)$$

$$x^* = r \cos \alpha \cos \theta - r \sin \alpha \sin \theta$$

$$y^* = r \sin \alpha \cos \theta + r \cos \alpha \sin \theta$$

$$x^* = x \cos \theta - y \sin \theta$$

$$y^* = x \sin \theta + y \cos \theta$$

$$p^* = \begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{bmatrix}$$

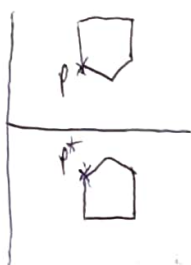
$$= \begin{bmatrix} x \\ y \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$p^* = [P] [T_r]$$

$[T_r] \Rightarrow$ Transformation matrix for rotation.

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3. Reflection / mirroring



about x axis,

$$p^* = \begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ -y \end{bmatrix}$$

If about y axis,

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -x \\ y \end{bmatrix}$$

In general, $[p^*] = [T_m] [p]$

$[T_m] =$ Transformation matrix for mirror.

4. Scaling



$$p = [x, y]$$

$$p^* = [x^*, y^*] = [s_x x, s_y y]$$

$$[p^*] = \begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$[p^*] = [T_s] [p]$$

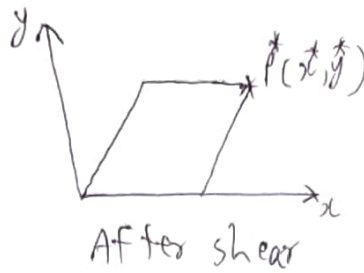
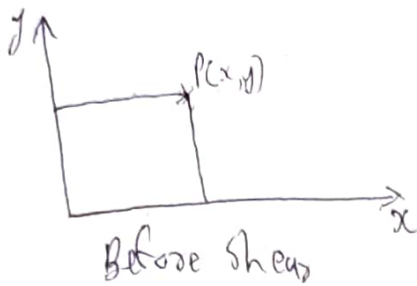
$[T_s] =$ Transformation matrix of scaling.

If $s_x = s_y \Rightarrow$ Uniform scaling

$s_x \neq s_y \Rightarrow$ Non uniform scaling.

5. Shear transformation

a. Shear about x axis

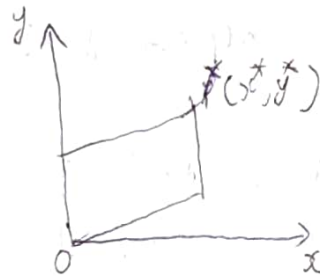
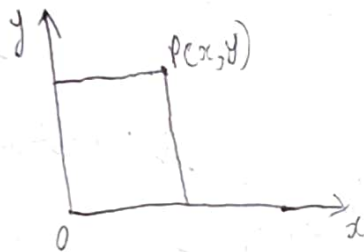


$$[P^*] = \begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} s_{hx} & 0 \\ s_{hx} & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$[P^*] = [T_{shx}] [P]$$

s_{hx} = Shearing along x-direction

b. Shear in y axis



$$[P^*] = \begin{bmatrix} x^* \\ y^* \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & s_{hy} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$[P^*] = [T_{shy}] [P]$$

s_{hy} = Shearing along y direction

NOTE: $P^* = [S][R][P]$
 $[S] \Rightarrow$ Scaling matrix
 $[R] \Rightarrow$ Rotation matrix
 } Rotation happens 1st, then scaling.

6. Composite/Concatenated transformation

It is the combination of 2 or more transformation such as translation, rotation, scaling etc. it will improve performance of

NOTE: Consider $P(x,y)$ is reoriented to $P^*(x^*, y^*)$
 First rotating, then scaling

7. Homogeneous transformation

The various rigid body transformation discussed so far represent translation, rotation, scaling, mirror & shear are

in the form of matrix multiplication. But translation takes form of vector addⁿ. This makes inconvenient to composite transformation. It is desirable to express all geometric transformation in the form of matrix multiplication only.

∴ Homo. trans. offer to represent both translation and rotation in a unified framework using homogeneous coordinates.

NOTE: ① In Homo. representation, an 'n' dimensional space is mapped to (n+1) dimensional space

② The 2D point is represented by $[x, y] \rightarrow [x, y, 1]$ ^{unity}

$$[P^*] = \begin{bmatrix} x^* \\ y^* \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & dx \\ 0 & 1 & dy \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \Rightarrow \begin{matrix} (x+dx) \\ (y+dy) \\ 1 \end{matrix} \begin{matrix} \nearrow \text{Translation} \\ \nearrow \end{matrix}$$

$$[T_t] = \begin{bmatrix} 1 & 0 & dx \\ 0 & 1 & dy \\ 0 & 0 & 1 \end{bmatrix}$$

$$[T_s] = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$[T_r] = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3D transformation

The transformation matrix will be 4×4 .

① Translation

$$\begin{bmatrix} x^* \\ y^* \\ z^* \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

② Scaling

$$\begin{bmatrix} x^* \\ y^* \\ z^* \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

③ Reflection

$$\begin{bmatrix} x^* \\ y^* \\ z^* \\ 1 \end{bmatrix} = \begin{bmatrix} \pm 1 & 0 & 0 & 0 \\ 0 & \pm 1 & 0 & 0 \\ 0 & 0 & \pm 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

④ Rotation about Z-axis (xy plane)

$$\begin{bmatrix} x^* \\ y^* \\ z^* \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta & 0 & 0 \\ \sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

b. Rotation about X-axis (yz plane)

$$\begin{bmatrix} x^* \\ y^* \\ z^* \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

c. Rotation about Y-axis (zx plane)

$$\begin{bmatrix} x^* \\ y^* \\ z^* \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 & \sin \theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin \theta & 0 & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- A line PQ with end points $P(2, 3)$, $Q(7, 8)$ is to be rotated about origin by 30° in c.w. determine coordinate of end point for rotated line.

NOTE: For c.w. direction, θ is -ve
For c.c.w direction, θ is +ve

$$P^* = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

$$Q^* = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$P^* = \begin{bmatrix} \cos(-30) & -\sin(-30) \\ \sin(-30) & \cos(-30) \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{3} + \frac{3}{2} \\ -1 + \frac{3\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} 3.232 \\ 1.598 \end{bmatrix}$$

$$Q^* = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3} \cdot 7}{2} + 4 \\ -\frac{7}{2} + 4\sqrt{3} \end{bmatrix} = \begin{bmatrix} 10.062 \\ 3.428 \end{bmatrix}$$

- Given a point $P(1, 3, -5)$. Find a transformed point P^* if P is translated by $d = 2\hat{i} + 3\hat{j} + 4\hat{k}$ and then rotated by 30° about Z axis

Trans. matrix for translation,

$$\begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$P_T^* = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ -5 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+2 \\ 3+3 \\ -5-4 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ -9 \\ 1 \end{bmatrix}$$

$$P_{TR}^* = \begin{bmatrix} \cos 30^\circ & -\sin 30^\circ & 0 & 0 \\ \sin 30^\circ & \cos 30^\circ & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \\ -9 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3} \cdot 3}{2} - 3 \\ \frac{3}{2} + \sqrt{3} \cdot 3 \\ -9 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.402 \\ 6.696 \\ -9 \\ 1 \end{bmatrix}$$

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Geometric Modelling

The user or designer constructs a geometrical model of an object on a CAD system by inputting the object data as required by the modelling technique via the user interface [GUI] provided by the software.

Software converts such data into mathematical representation which stores in the model data base.

Wire frame modelling

The image assumes the appearance of a frame constructed out of wire hence named as wireframe model.

↳ Wireframe model is a simplest model that can be represented mathematically of an object.

↳ Edge representation of object.

↳ Wireframe model consists of points, lines, Arcs, circles, conics and curves.

↳ Simplicity to construct, doesn't require computation time and memory.

↳ Considered as a natural extension of traditional method of drafting - doesn't require extensive training.